MAANAS RASTOGI

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EDUCATION

Bachelor of Science, Electrical/Computer Engineering **Expected May 2028**
Integrated Circuits and Systems *GPA: 3.7/4.0*
The University of Texas at Austin

Relevant Coursework: Digital Logic Design, Circuit Theory, Data Structures, Solid State Devices, Electromagnetism, Embedded Systems, Linear Systems and Signals

Independent study: CMOS and VLSI design

TECHNICAL EXPERIENCE

Validation/Design Intern | *Ironlattice* *August 2026 - Present*

- Review research papers on ferroelectric devices, bulk crystal growth, and phase stabilization to guide 1-FeFET superlattice memory modeling for a 40 nm node roadmap.
- Analyze equivalent oxide thickness (EOT), dielectric constants, and interfacial capacitance to assess gate-stack scaling and thickness trade-offs for $4F^2$ memory-cell architectures.
- Develop initial SPICE simulations to explore design targets of 10^{10} endurance cycles, 10 ns switching, and 30+ analog states; plan device-physics simulations while arranging software access.

Racing Team Electronics Lead | *Longhorn Baja Racing* *August 2025 - Present*

- Lead 10+ members through vehicle electrical integration, design reviews, and race-day telemetry validation; organize workshops on Altium Designer, circuit theory, and PCB design.
- Design 20-gauge power distribution and custom dashboard hardware for a Baja SAE race car; integrate 7+ onboard sensors into a virtual driving coach and vehicle-monitoring system.
- Develop 3D spatial-mapping and vehicle-control algorithms targeting under 50 ms processing latency, translating mapped terrain into steering and throttle inputs.

Incoming Undergraduate Researcher | *Banerjee Lab, UT Austin MRC* *October 2026 - Present*

- Planned research (10-20 hours/week): investigate yttrium-doped HfO_2 and ferroelectric superlattices, studying dopant concentration, annealing, and orthorhombic phase stabilization on Ge/Si.
- Planned fabrication: train in atomic layer deposition, sputtering, photolithography, etching, and rapid thermal annealing to build 3 test-device types: MFM capacitors, MFIS stacks, and FeFETs.
- Planned characterization: measure polarization switching, interface trap density, gate leakage, and endurance; use ellipsometry, X-ray diffraction, and atomic force microscopy to assess thickness and structure.

Quantum Computing Undergraduate Researcher | *UT Austin* *October 2025 - December 2025*

- Built a simulation pipeline using Stim, sinter, and PyMatching to model surface-code stabilizer circuits across modular, chiplet-based quantum processors.
- Ran 100,000+ Monte Carlo simulations to quantify logical error scaling and evaluate the effect of 1% inter-chip network noise on fault tolerance.

Electrical Engineer | *UT Austin Robotics & Automation Society* *August 2024 - August 2025*

- Reworked the supercapacitor bank and implemented a new current-limiting circuit to regulate current draw and improve power delivery, increasing output from 12 W to 90 W and extending runtime by 20%.
- Redesigned the main control board around a high-efficiency charging circuit, increasing charging current to 6 A and cutting charge time by 75% while making the board easier to access and maintain.

PROJECTS

VLSI Design - RISC-V Processor

- Design a custom 32-bit RISC-V processor in Verilog, using Vivado for simulation and synthesis.
- Develop an OpenROAD physical-design flow targeting a 50 MHz clock frequency and a 15% reduction in area overhead.

Motorcycle Helmet HUD

- Built a Raspberry Pi Android Auto head unit with a 5-inch touchscreen and input latency below 100 ms.
- Develop Python controls and a prism-based optical display targeting 60 Hz ride data, a 20-degree field of view, and 85% visor transparency.
- Plan to use a lens and a see-through reflector to bounce the screen image toward the rider's eye, so ride information appears over their view of the road.

SKILLS

Programming: *Python*, *C/C++*, *Java*, *Assembly*, *Verilog*
Tools: *SPICE*, *Altium Designer*, *Vivado*, *OpenROAD*, *Stim*, *sinter*, *PyMatching*
Technical areas: *PCB design*, *embedded systems*, *circuit simulation*, *quantum error correction*, *semiconductor literature review*